

## Assignment 3

30 Marks

Deadline: Dec 27, 2025 (Saturday)

### Question - 1:

Why is a bias added to the actual exponent in the IEEE 754 floating-point representation, and how does this affect the encoding of both positive and negative exponents? [5]

### Question - 2:

How does optimized multiplication improve efficiency and performance compared to traditional long multiplication, especially in terms of speed and computational complexity? [5]

### Question - 3:

Multiply  $(10110)_2$  with  $(101)_2$  using optimized multiplier, you are required to show the whole simulation. [10]

### Question - 4:

Perform the following multiplication using 24 bit IEEE-754 floating-point representation where the size of the fraction field is 14 bits. Check if the result has overflow or underflow.

$$(0.0101)_2 \times 2^{79} \text{ and } (10010.0101)_2 \times 2^{87}$$

Note: Consider 8 decimal digits while converting from decimal to binary for the following questions. [10]

### Question - 5: [Ungraded]

Convert the following code to RISC-V assembly language.

```
float a = fp_array [0];
```

```
float b = fp_array [1];  
int c = 0;  
if (a < b) {  
    c = (int) a;  
} else {  
    c = (int) b;  
}
```

Where fp\_array is a floating point word array base address of which is stored in x13.